HW 7 Silvaco TCAD Exercise (due on 5/11/2026)

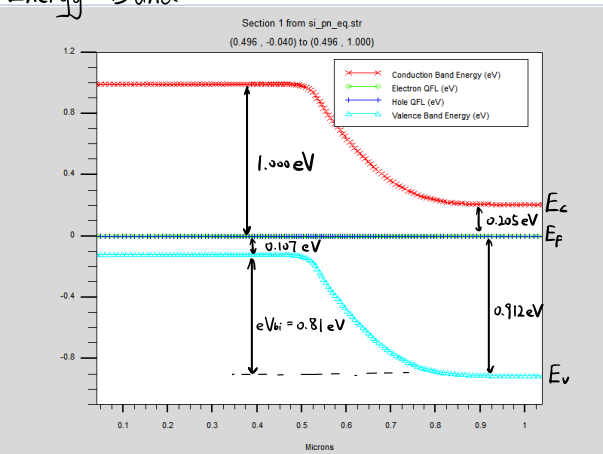
Simulation and Theoretical Analysis of a Silicon PN Junction using Silvaco Atlas

Objective: To design a Silicon PN junction using the Silvaco Atlas TCAD tool, analyze its electrostatic behavior at equilibrium and under reverse bias, and evaluate the validity of the Depletion Region Approximation (DRA) by comparing simulated data with theoretical models.

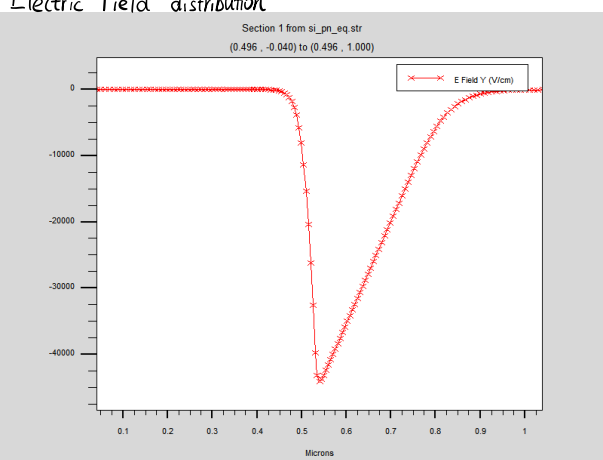
Assume an abrupt junction with doping concentrations (e.g., $N_A = 1\times10^{17}cm^{-3}$ and $N_D = 1\times10^{16}cm^{-3}$).

1) Develop a Silvaco Atlas input file (.in) to construct a Silicon PN junction. Perform a zero-bias simulation and extract the Energy Band Diagram (showing Ec ,Ev , and Ef), depletion region width, and electric field distribution across the junction
2) Using the Depletion Region Approximation (DRA), manually calculate the analytical values for the device parameters defined in Task 1).
3) Compare simulated and theoretical values and schematic. Explain discrepancies.
4) Apply reverse biases V_R = 5V, 10V to the PN junction, generate schematic diagrams of its energy band structure, depletion region width, and electric field distribution within the device under different biases. Explain discrepancies under different bias

1) Energy Band



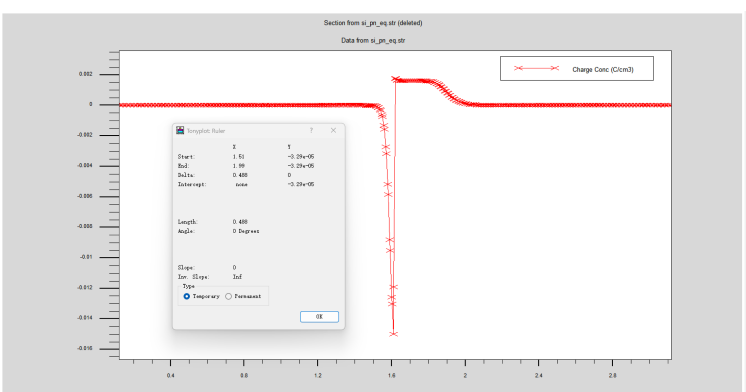
Electric Field distribution



Maximum Electric Field:

$$E_{max} = 4.4 \times 10^4 \text{ V/cm}$$

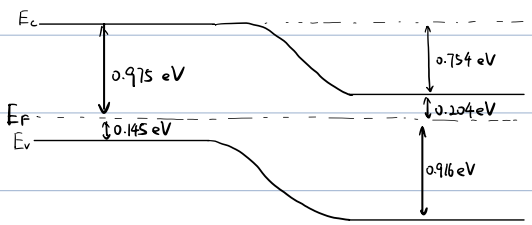
$x_p = 0.097 \mu m$. $x_n = 0.391 \mu m$



Depletion Region width : 0.488 μm

2) Region P: $E_c - E_f = 0.975\ eV$. $E_f - E_v = 0.145\ eV$

Region n: $E_c - E_f = 0.204\ eV$. $E_f - E_v = 0.916\ eV$



$$V_{bi} = \frac{kT}{e} \cdot \ln \frac{N_A \cdot N_D}{n_i^2}$$

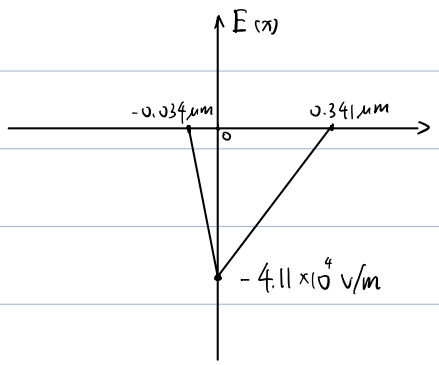
$$E_{max} = \sqrt{\frac{2eV_{bi}}{\varepsilon_0 \varepsilon_r}\left(\frac{N_A N_D}{N_A + N_D}\right)} \quad . \quad \varepsilon_r = 11.7$$

$$x_n = \sqrt{\frac{2\varepsilon_0 \varepsilon_r V_{bi}}{e}\left(\frac{N_D}{N_A(N_A+N_D)}\right)} . \quad x_p = \sqrt{\frac{2\varepsilon_0 \varepsilon_r V_{bi}}{e}\left(\frac{N_A}{N_D(N_A+N_D)}\right)} .$$

$$W = x_p + x_n$$

$\Rightarrow$ Theoretical value: $V_{bi} = 0.754V$. $E_{max} = 4.11 \times 10^4\ V/cm$. $x_p = 0.034\mu m$. $x_n = 0.341\ \mu m$.

$W = 0.375\ \mu m$



3) Theoretical value

Region P: $E_c - E_F = 0.975\ eV$. $E_F - E_v = 0.145\ eV$

Region n: $E_c - E_F = 0.204\ eV$. $E_F - E_v = 0.916\ eV$

$E_{max} = 4.11 \times 10^4\ V/cm$

$x_p = 0.034 \mu m$.

$x_n = 0.341 \mu m$.

$W = 0.375 \mu m$

Simulated value

Region P: $E_c - E_F = 1.000\ eV$. $E_F - E_v = 0.107\ eV$

Region n: $E_c - E_F = 0.205\ eV$. $E_F - E_v = 0.912\ eV$

$E_{max} = 4.40 \times 10^4\ V/cm$

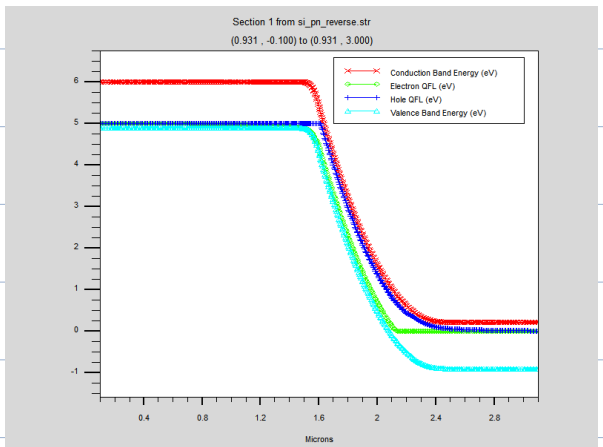
$x_p = 0.097 \mu m$.

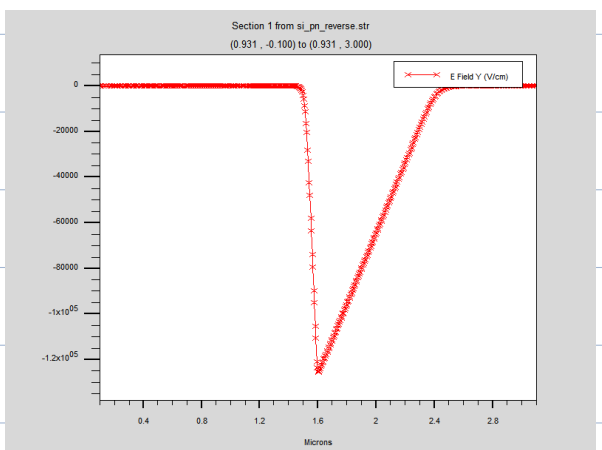
$x_n = 0.391 \mu m$.

$W = 0.488 \mu m$

- The DRA assumes an abrupt junction with a sharp depletion boundary, where free carriers drop to zero instantly. In reality (simulation), there is a smooth transition region at the depletion edges.
- The DRA uses Boltzmann statics while the simulation applys Fermi-Dirac statistics.
- The DRA assumes neutral regions have zero electric field. In simulation, weak electric fields exist near the depletion edges to balance carrier diffusion.

4) $V_R = 5\ V$

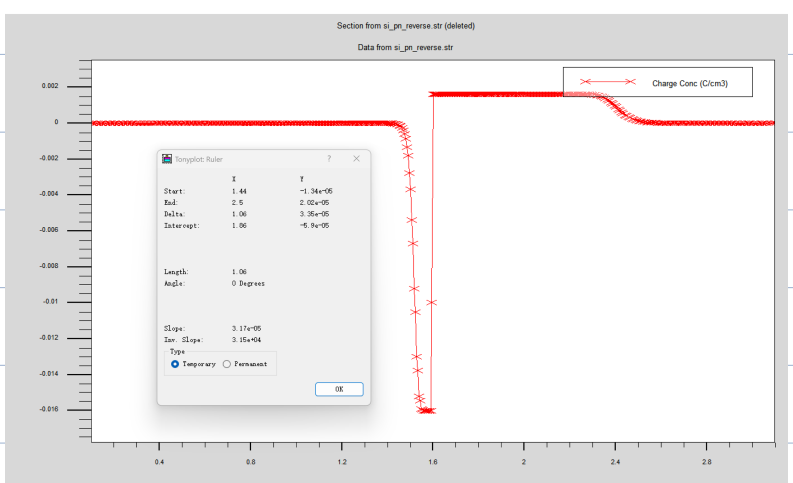




Maximum Electric Field:

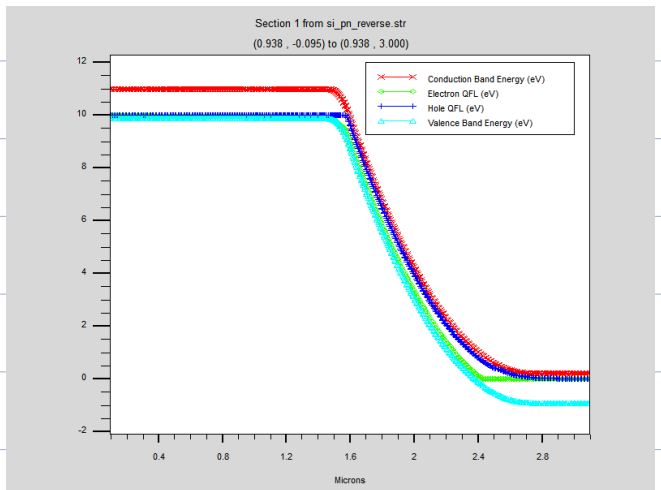
$$E_{max} = 1.26 \times 10^5\ V/cm$$

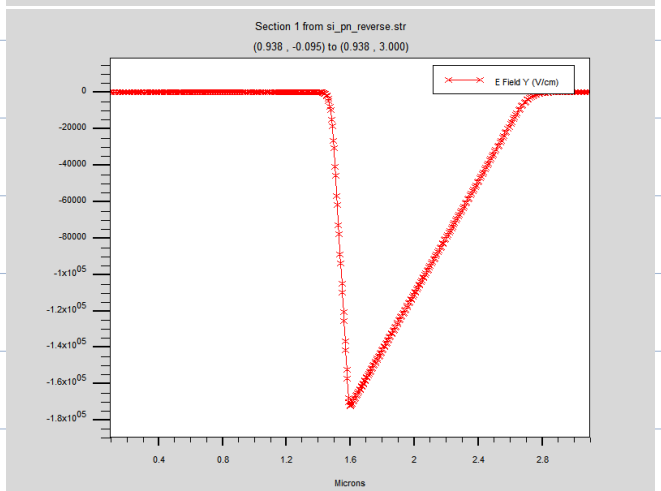
$x_p = 0.127\ \mu m$. $x_n = 0.933\ \mu m$



Depletion Region width: 1.06 μm

$V_R = 10V$

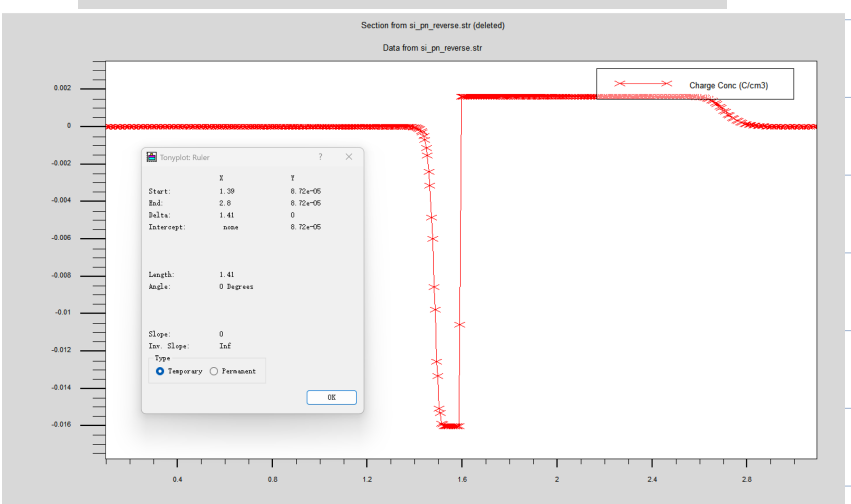




Maximum Electric Field:

$E_{max} = 1.71 \times 10^5$ V/cm

$x_p = 0.166\,\mu m$. $x_n = 1.244\,\mu m$



Depletion Region width: $1.41\,\mu m$

- When applying reverse biases voltage, the depletion region width becomes large, the relative influence of the transition region decreases, the error in the DRA becomes smaller.
- The simulation includes reverse leakage current, which slightly modifies the space-charge distribution, while the DRA completely ignores the presence of current.
- The DRA ignores carrier velocity saturation. In simulation, it will cause mobility decreases under high electric fields, which alters the space charge distribution and affects the depletion region width.